

Model Paper - Database Management Systems

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Q1 (20 marks)

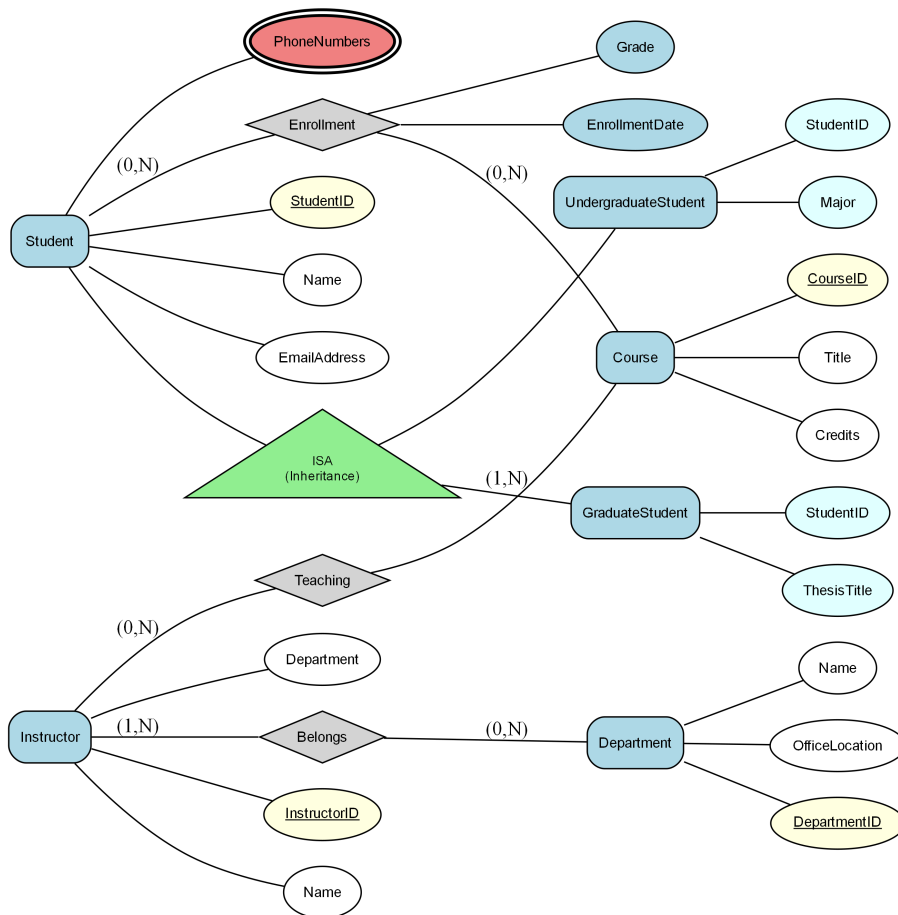
Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)
- (0,1) indicates optional participation (zero or one)
- (0,N) or (0,*) indicates optional many participation (zero or many)

In a university database system, the entities include Student (StudentID, Name, EmailAddress, PhoneNumbers), Course (CourseID, Title, Credits), Instructor (InstructorID, Name, Department), and Department (DepartmentID, Name, OfficeLocation). Students enroll in Courses through an Enrollment relationship, which has attributes EnrollmentDate and Grade. Courses are taught by Instructors, and each Instructor belongs to a Department. There are subtypes for Student such as GraduateStudent (ThesisTitle) and UndergraduateStudent (Major), both inheriting attributes from Student. A Student can be related to many Courses, and a Course can be related to many Students through the 'Enrollment' relationship (Student participation is optional, Course participation is optional). A Instructor can be related to many Courses through the 'Teaching' relationship (Instructor participation is optional). Many Instructors can be related to one Department through the 'Belongs' relationship (Department participation is optional). Draw the EER diagram

Figure: EER



a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17 marks)

b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3 marks)

Q2 (15 marks)

Consider a relation $R(A, B, C, D, E)$ with the following set of functional dependencies F over R :
 $F = \{AB \rightarrow C, B \rightarrow D, C \rightarrow E, A \rightarrow BD\}$. Analyze this relation to determine its keys, normal form compliance, and any necessary decompositions.

a) Find all the keys in relation R using the attribute closure method. (4 marks)

b) Is R in 3NF? Give reasons for your conclusion. (3 marks)

c) Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations. (8 marks)

Q3 (25 marks)

A university is developing a database to manage its course offerings and student enrollments. The database will include tables for Courses, Students, and Enrollments. Each course will have attributes such as CourseID, CourseName, and Credits. Students will have attributes like StudentID, Name, and Major. Enrollments will link Students to Courses and have attributes like EnrollmentID and Grade. The university aims to maintain data integrity and optimize queries using various constraints and

statements.

a) In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement justifying your answer. (2 marks)

b) Code Segment:

```
```sql
CREATE TABLE Courses (
 CoursesID CHAR(10) PRIMARY KEY,
 Name VARCHAR(50) NOT NULL,
 NIC CHAR(10) UNIQUE,
 Age INT,
 GPA FLOAT,
 StudentsID VARCHAR(10) DEFAULT 'D001',
 EnrollmentDate DATE DEFAULT GETDATE(),
 CONSTRAINT courses_students_fk FOREIGN KEY (StudentsID)
 REFERENCES Students(StudentsID)
 ON DELETE SET DEFAULT
 ON UPDATE CASCADE,
 CONSTRAINT gpa_check CHECK (GPA >= 0.0 AND GPA <= 4.0),
 CONSTRAINT age_check CHECK (Age >= 18 AND Age <= 100)
);
```
```

Consider the SQL code provided below. Which type of statements is used in the code segment shown above? Briefly explain when this type of statement will be used. (2 marks)

c) Briefly explain the different types of integrity constraints that can be implemented in the above database schema. (2 marks)

d) Briefly explain how cascading updates and deletes can enhance referential integrity in the context of the Enrollments table. (3 marks)

e) A student named Alex is enrolling in a new course.. Michael is a data entry operator responsible for inputting data into the system.. Nathan is a database developer responsible for designing and implementing database schemas.. Emily is a junior database administrator managing specific databases within the system.. Sarah is the senior database administrator tasked with overseeing the entire database system's creation and maintenance.. Write a T-SQL statement to insert a new record into the Enrollments table capturing Alex's enrollment details for a specific CourseID.

- i. Write a T-SQL statement to create a login to Sarah with windows authentication. (2 marks)
- ii. Provide Sarah with the necessary permissions to handle all administrative tasks within the server using a fixed server role. (3 marks)
- iii. Assuming Emily's username is 'emily.e', write T-SQL statements to assign her the responsibility of managing the transactions database using a user-defined server role. (5 marks)
- iv. Assuming Nathan's login name is 'nathan.n', write T-SQL statements to grant him permission to create tables and databases. (3 marks)
- v. Assuming Michael's login name is 'michael.m', write T-SQL statements to allow him to perform data entry

tasks on the client accounts and transactions databases. (3 marks)

Q4 (40 marks)

Consider the following schema of a database designed for an E-commerce platform: Product (productId: int, productName: varchar(100), price: real, stockQuantity: int) Order (orderId: int, customerId: int, orderDate: date, totalAmount: real) Customer (customerId: int, customerName: varchar(100), email: varchar(50)) OrderDetails (orderDetailId: int, orderId: int, productId: int, quantity: int, subTotal: real). The 'Product' table stores details about products available for sale, including a unique product ID, name, price, and the quantity in stock. The 'Order' table contains information about customer orders, including a unique order ID, the customer who placed it, the date of the order, and the total amount. The 'Customer' table maintains customer information with unique customer IDs, names, and email addresses. The 'OrderDetails' table captures details about the products within an order, including the quantities ordered and their respective subtotals. The 'Customer' table manages details about customers, including unique customer ID, name, email, and address. The 'OrderDetails' table contains order records, tracking customer orders with order ID, customer ID, order date, and total amount.

a) Write SQL Queries to perform the following: i. Find the customer name and email for all customers who have placed an order on '2023-10-05'.

- i. Find the customer name and email for all customers who have placed an order on '2023-10-05'. (4 marks)
- ii. Find the customer who has the highest total totalamount compared to all other customers. Find the customerId and customerName. (6 marks)
- iii. Find the productName and customerName from Product, Customer, and Order for currently active records. (7 marks)

b) b) Create a function to calculate the total product amount for a given customer. This function should account for product with a late penalty of 10% applied to those marked as 'overdue' payment status. (11 marks)

c) c) Create a trigger that automatically updates the 'totalAmount' column in the 'Order' table whenever a new order record is added, or an existing order record is updated. The 'totalAmount' column should contain the total product amount for each customer. Ensure the trigger adjusts the 'totalAmount' column accurately to reflect any partial payments made by customer. The customer pays only 50% of the total product amount as an initial payment when the payment status is marked as 'Partial'. (12 marks)